Rafael Ngai

Software Developer

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Full right to work in the UK (no sponsorship needed)

Timetable live, ~100 users	**Student Tracker** 129 tests	**GorB** group gym app	**Website watchdog** daily checks	**FitTrack** offline PWA

Profile

Software developer who builds and runs the systems a two-site school depends on: a live timetable and rota platform used by about 100 people, and a learning tracker backed by 129 automated tests.

I build with AI tools under a written process: plan the whole project, write the document first, choose the right tools, agree how it will look, prototype, test, then ship. BSc in Virtual and Augmented Reality, with a background in C# and Unity.

Experience

Support Worker (Game Design), Overworld AMP

May 2025 – present

I support and teach students, and I am the school's in-house developer. Since September 2026 I have protected weekly time for development, agreed with management.

Timetable and rota platform (live). Built and run the system both sites plan their day from: a week editor, a published read-only rota, and a front-desk board for arrivals and pick-ups. Vanilla JavaScript, Node.js and a per-record document store.

- Used by about 100 people; in October 2026, 10 staff made over 31,000 changes, across about 570 sessions a week.
- Replaced a ready-made rota product the school would otherwise have bought, and removed manual spreadsheet admin.
- Pupils' home addresses are released by the server only to staff on that journey, and every read is logged. Record versions stop one person's save from overwriting another's.

Student Tracker (in progress). A Moodle-style learning platform that links 718 course units (3,068 criteria across 21 qualifications), with hand-in, marking and verification.

- Every change is a named, all-or-nothing action; each person's browser receives only the records they may see.
- 129 automated tests and 12 written architecture decision records.

Teaching. Game Design, C# in Unity, Python, website design and music theory, including for students with special educational needs. Built a Node.js pipeline (pptxgenjs) that generates my lesson slide decks.

Earlier experience

Freelance online marketing 2023 – 2025
Freelance online marketing work, alongside personal projects in Unity and Unreal Engine 5.

Documentation Clerk, DB Schenker 2018 – 2019

Research Assistant, The Chinese University of Hong Kong 2017 – 2018

Selected projects

GorB: a group gym check-in app on Supabase (PostgreSQL, Auth, Storage, Edge Functions) with Web Push. Row-level security on all 17 tables, server-stamped check-ins, photos stripped of location and deleted after 7 days, migrations deployed by GitHub Actions. Built in 2 days.

Website watchdog: Node.js, Playwright and axe-core on a daily GitHub Actions run. Checks 5 sites on desktop and phone for uptime, errors, broken links and accessibility; raises and closes GitHub Issues itself.

Portfolio site: a page run by a small timetable engine you can rearrange; scores 98–100 on the watchdog.

FitTrack: an offline-first fitness app (PWA) in vanilla JavaScript with Three.js.

VR Driving Test Simulator (university, group leader): Unity and C#, with automatic exam scoring. Also Farm Frenzy in Unreal Engine 5 (C++).

Skills

Languages	JavaScript, TypeScript, C#, SQL (PostgreSQL), Python, HTML, CSS, C++
Back end and data	Node.js, Supabase, row-level security, REST APIs, document stores
Front end	Accessible responsive UI (WCAG), progressive web apps, Three.js
Testing and delivery	Automated tests, Playwright, GitHub Actions CI/CD, Git, Perforce, Jira
AI-assisted development	Claude Code with custom skills; plan, document, prototype, test
Engines and tools	Unity, Unreal Engine 5, 3ds Max, Photoshop, Illustrator

Education

BSc (Ordinary) in Virtual and Augmented Reality, with Merit, University of Portsmouth 2022 – 2023

Higher Diploma in Multimedia and Virtual Reality, Hong Kong IVE (final project: Grade A) 2019 – 2021

> "Working alongside Raf while he built the rota system was a joy. He was open to all feedback, and his communication was excellent throughout."
>
> Andrew, Supervisor Subject Lead

Languages: English, Cantonese, Mandarin **References:** available on request